

04 — Data Model & Logic

PostgreSQL/Prisma entities, constraints, indexes and business rules

CampusFlow • Production Specification • v1.0 • October 2026

1. Core Entities

Department: UUID id, name/code unique; MCA and MBA.

User: UUID id, name, normalized unique email, passwordHash, role, departmentId, year FY/SY, division A/B, batch F1/F2/F3, status, mustChangePassword, timestamps.

Notice: title, content, category EXAM/ACADEMIC/EVENT/GENERAL, priority URGENT/HIGH/NORMAL, status DRAFT/PUBLISHED/ARCHIVED, publishedAt, authorId, sourceUrl, rawFileHash unique.

NoticeAudience: noticeId + department/year/division/batch.

NoticeSummary: noticeId unique, whatChanged, whoAffected, requiredAction, deadline, rawAIResponse.

Event: title, description, date, venue, capacity, registrationDeadline, status, createdById.

Registration: eventId, userId, status, registeredAt; unique eventId+userId.

Deadline: title, description, dueAt, type EXAM/SUBMISSION, subject, status, createdById.

Notification: userId, type, title, body, readAt, createdAt.

Attachment: storageKey, filename, mimeType, size, noticeId.

AuditLog: actorId, action, entityType, entityId, metadata JSON, createdAt; append-only.

2. Prisma Core Schema

```
enum Role { STUDENT FACULTY ADMIN }
enum UserStatus { ACTIVE INACTIVE }
enum Year { FY SY }
enum Division { A B }
enum Batch { F1 F2 F3 }
enum NoticeCategory { EXAM ACADEMIC EVENT GENERAL }
enum NoticePriority { URGENT HIGH NORMAL }
enum NoticeStatus { DRAFT PUBLISHED ARCHIVED }
enum EventStatus { DRAFT PUBLISHED ARCHIVED }
enum RegistrationStatus { CONFIRMED CANCELLED }
enum DeadlineType { EXAM SUBMISSION }
```

```
model Department { id String @id @default(uuid()) name String code String @unique }
model User { id String @id @default(uuid()) name String email String @unique passwordHash String role Role @default(STUDENT) status UserStatus }
model Notice { id String @id @default(uuid()) title String content String category NoticeCategory priority NoticePriority @default(NORMAL) status NoticeStatus }
model NoticeSummary { noticeId String @id whatChanged String whoAffected String requiredAction String? deadline DateTime? rawAIResponse Json? }
model Event { id String @id @default(uuid()) title String description String date DateTime venue String capacity Int? registrationDeadline DateTime }
model Registration { id String @id @default(uuid()) eventId String userId String status RegistrationStatus @default(CONFIRMED) registeredAt DateTime }
```

3. Business Rules

- Email normalization: remove whitespace + lowercase.
- Registration uniqueness enforced by database.
- Capacity checked transactionally.
- Published content retains author and source.
- Important records are archived/soft-deleted.
- Audit logs are append-only.
- Notifications are downstream of successful source transactions.

4. Priority Logic

Urgent official information → Urgent.

Action due within 24h → Due today.

Action due within 3 days → Due soon.

Important but not immediate → Important.

General information → Normal.

5. Supplied Academic Calendar Seed

Display of Time Table: 05/09/2026.

CIE-I: 29/10/2026–06/11/2026.

CIE-II: 28/12/2026–02/01/2027.

SEE: 11/01/2027.

6. Indexes

Notice publishedAt/status/category.

Event date/status.

Deadline dueAt/status.

Notification userId/readAt/createdAt.

Registration eventId and unique eventId/userId.

Audit createdAt and entity identifiers.